Digital Information Literacy & Computer Application Jobs

Group 3
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Introduction

Information Technology (IT) is one of the fastestgrowing fields in the United States, offering a wide range of career paths that require different levels of education, technical skills, and professional responsibilities. As part of our exploration for ITE 152, We reviewed resources such as the NVCC Library (*Northern Virginia Community College Library, 2024*). The Department of Labor's Occupational Outlook Handbook, (*U.S. Bureau of Labor Statistics, 2024*) and several IT career guides. During this research process, We focused on careers that matched our interests in problemsolving, technology, and digital literacy. After comparing several options, We narrowed our choices to two career paths: Cybersecurity Analyst and Computer Support Specialist. These careers stood out because they combine technical skills with realworld problem solving and offer strong job growth.

Career Path 1: Cybersecurity Analyst

Job Title

Cybersecurity Analyst (also called Information Security Analyst)

Job Description

A Cybersecurity Analyst protects an organization's computer networks, systems, and data from cyberattacks. Daily tasks include monitoring network activity, identifying security breaches, installing protective software, conducting vulnerability tests, and responding to incidents. Analysts must understand firewalls, encryption, intrusion detection systems, and risk management. Strong analytical thinking, attention to detail, and problemsolving skills are essential for this role. Cybersecurity Analysts often work with IT teams to create security policies, train employees on safe digital practices, and ensure compliance with federal or industry regulations.

Education and Training Requirements

Most cybersecurity positions require a bachelor's degree in cybersecurity, information technology, or computer science. Some employers accept an associate degree combined with certifications. Common certifications include:

- CompTIA Security+
- Certified Ethical Hacker (CEH)
- Certified Information Systems Security Professional (CISSP)

Many employers also require ongoing training because cybersecurity threats constantly evolve. *(U.S. Bureau of Labor Statistics, 2024)*.

Salary

According to the Occupational Outlook Handbook, Cybersecurity Analysts earn a median annual salary of around $120,000. *(U.S. Bureau of Labor Statistics, 2024)*. Higher education and advanced certifications significantly increase earning potential.

Job Benefits & Drawbacks

Benefits:

- High salary and strong job security
- Growing demand due to increasing cyber threats
- Opportunities to work in government, healthcare, finance, and private industry
- Engaging work that involves solving complex problems

Drawbacks:

- High stress during security breaches
- Requires constant learning and staying updated
- May involve long hours or emergency response situations

Opportunities for Advancement

Cybersecurity Analysts can advance to roles such as Cybersecurity Manager, Penetration Tester, Security Architect, or Chief Information Security Officer (CISO). Advancement typically requires additional certifications, leadership skills, and experience. Continuing education is essential because new threats and technologies appear regularly.

A Typical Day in the Life

A typical day includes reviewing security logs, monitoring alerts, analyzing suspicious activity, updating security software, and meeting with IT teams to discuss risks. Analysts may also conduct penetration tests or respond to incidents. Some days are routine, while others involve urgent action to stop or investigate cyberattacks.

Summary of Appeal

Cybersecurity appeals to us because it is fastpaced, intellectually challenging, and important for protecting organizations. The high salary and strong job growth are also attractive. Less appealing aspects include stress and constant need for continuing education.

Career Path 2: Computer Support Specialist

Job Title

Computer Support Specialist (also called IT Support Technician or Help Desk Technician)

Job Description

Computer Support Specialists assist users with technical problems involving hardware, software, networks, and digital tools. *(U.S. Bureau of Labor Statistics, 2024)*. They troubleshoot issues, answer questions, install software, configure devices, and help maintain computer systems. This role requires strong communication skills, patience, and basic technical knowledge. Support Specialists often work in offices, call centers, schools, or government agencies.

Education and Training Requirements

Many positions require only a high school diploma or associate degree, making this an accessible entrylevel IT career. Helpful certifications include:

- CompTIA A+
- CompTIA Network+
- Microsoft 365 Certified: Fundamentals (No license is required)

Salary

The Occupational Outlook Handbook lists the median salary for Computer Support Specialists at around $60,000 per year, *(U.S. Bureau of Labor Statistics, 2024)*. Depending on experience and industry. Higher education and certifications can increase pay.

Job Benefits & Drawbacks

Benefits:

- Easy entry into the IT field
- Opportunities to learn handson skills
- Flexible work environments
- Strong demand in schools, businesses, and government

Drawbacks:

- Lower salary compared to other IT careers
- Repetitive tasks
- Dealing with frustrated customers
- Limited autonomy in some workplaces

Opportunities for Advancement

Support Specialists can advance to roles such as Network Administrator, Systems Administrator, or Cybersecurity Technician. Advancement usually requires additional certifications and experience. Continuing education is helpful but not always required.

A Typical Day in Life

A typical day includes responding to help desk tickets, troubleshooting computer issues, installing updates, assisting employees with login problems, and maintaining hardware. The work is steady and predictable, with occasional urgent issues.

Summary of Appeal

This career appeals to us because it offers a clear entry point into IT and provides handson experience with computers. It is a good starting job for building skills. Less appealing aspects include the lower salary and the customerservice challenges.

Conclusion

Both Cybersecurity Analyst and Computer Support Specialist careers offer valuable opportunities in the IT field. The Computer Support Specialist role provides an accessible entry point with practical experience, while the Cybersecurity Analyst role offers higher pay, more responsibility, and longterm growth. After comparing both paths, We're most interested in becoming a Cybersecurity Analyst because it aligns with our interest in problemsolving, digital protection, and longterm career advancement. Although it requires more education and ongoing training, the benefits outweigh the drawbacks and make it a strong career choice.

Annotated Bibliography

Library, N. V. (2024). *Information Technology Career Resources*. Retrieved from https://www.nvcc.edu/academics/library/index.html

This source provided an overview of information technology careers and research tools available through Northern Virginia Community College. It helped us identify different career paths within the IT field and understand the education and skills required for those careers. The resource was useful during the early stages of our research because it pointed us toward reliable career information and professional resources. We used this source to explore career options and narrow our focus to Cybersecurity Analyst and Computer Support Specialist positions.

Statistics, U. B. (2024). *Computer Support Specialists*. Retrieved from https://www.bls.gov/ooh/computer-and-information-technology/computer-support-specialists.htm

This source provided detailed information about the duties, education requirements, salary, job outlook, and work environment of Computer Support Specialists. Because it is published by the U.S. Bureau of Labor Statistics, it is a highly reliable government source. We used the information to describe the responsibilities of Computer Support Specialists and evaluate the benefits and drawbacks of the career. The salary and employment outlook data helped us compare this career path with cybersecurity.

Statistics, U. B. (2024). *Information Security Analysts*. Retrieved from https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm

This source explained the role of Information Security Analysts, including job responsibilities, education requirements, certifications, salary, and projected job growth. The source was especially valuable because it provided current labor market information from a trusted government agency. We used this information to support our discussion of Cybersecurity Analysts and explain why the field appeals to us. The strong job outlook and high earning potential identified in this source influenced our conclusion that cybersecurity is our preferred career path.